Food Delivery Application (SDSE)

1. Introduction

The SDSE Food Delivery Application is a full-stack web platform designed to streamline the process of online food ordering. It provides a seamless experience for customers to browse menus, manage carts, and place orders, while also offering a dedicated administrative dashboard for restaurant operators to manage food items and track orders efficiently. This project demonstrates strong application of modern web development practices, including component-based architecture, RESTful APIs, and secure authentication mechanisms.

2. System Architecture

The system follows a decoupled client-server architecture, ensuring scalability and maintainability. The backend is responsible for handling business logic, authentication, and database interactions, while the frontend focuses on delivering a responsive and interactive user interface. The Model-View-Controller (MVC) pattern is implemented on the server side to separate concerns effectively. This separation ensures that updates to one part of the system do not negatively impact others, making the application easier to scale and debug.

3. System Design Diagrams

The following diagrams illustrate the internal working and structure of the application. These include sequence diagrams representing user interactions, and component diagrams showing how different parts of the system communicate and share data.

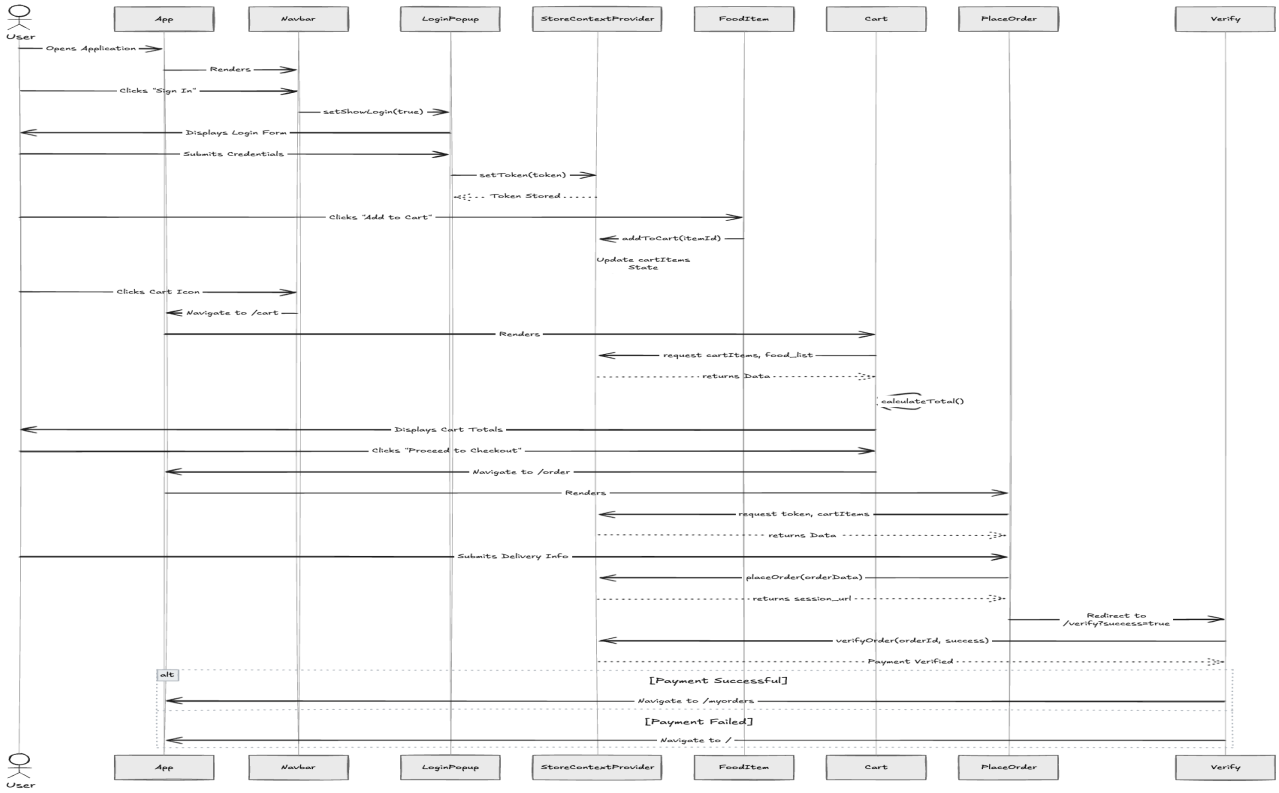

Figure 1: Sequence Diagram showing user flow from login to order placement and payment verification.

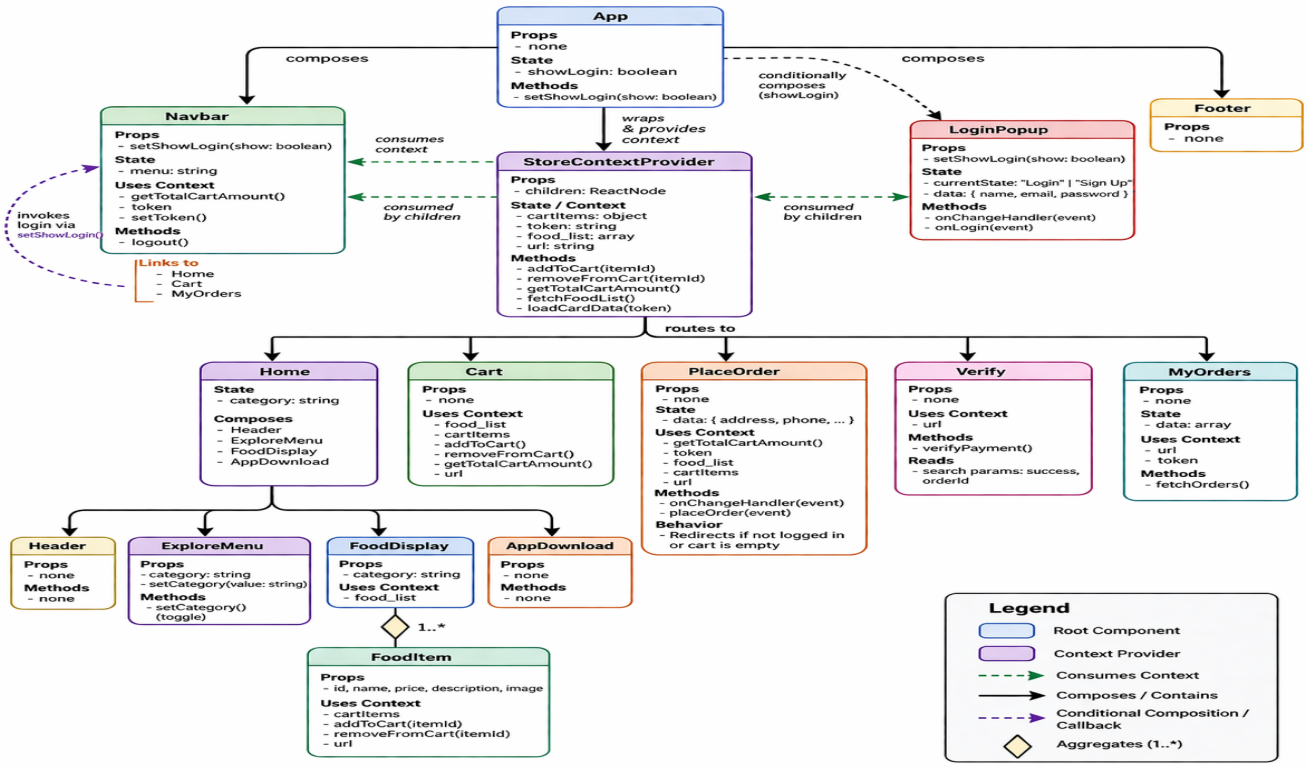

Figure 2: Component Diagram illustrating React component hierarchy and context usage.

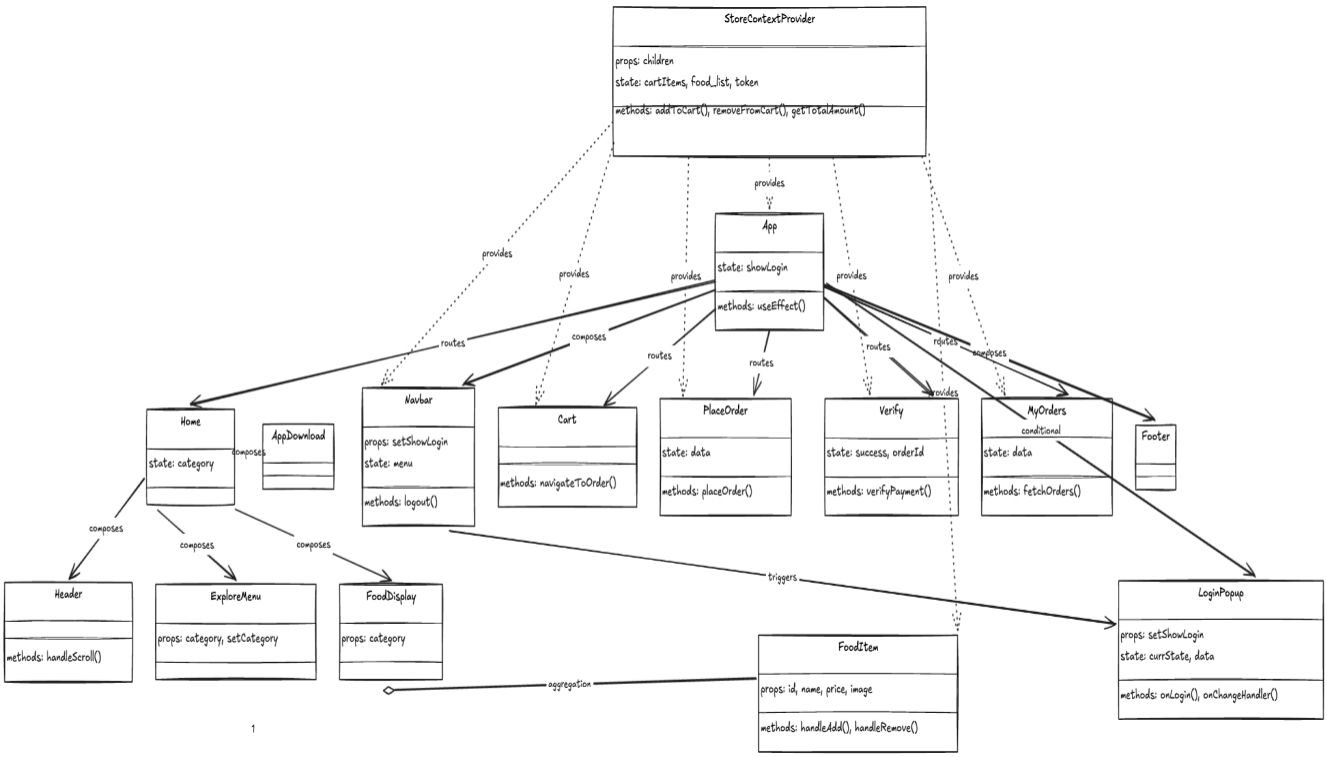

Figure 3: Simplified Architecture Diagram representing routing and component interactions.

4. Key Features

Customer Module:
- Secure authentication using JWT ensuring safe login sessions.
- Dynamic menu browsing with real-time data fetched from backend APIs.
- Cart management system allowing users to add, remove, and update item quantities.
- Seamless checkout process with order placement and payment verification.

Admin Module:
- Role-based access control to restrict sensitive operations.
- CRUD operations for managing food items and updating menu dynamically.
- Order tracking system enabling administrators to monitor and update order status.

5. Software Engineering Principles

- **Separation of Concerns:** Frontend and backend are separated into independent modules.
- **Encapsulation:** Business logic is hidden within controllers and context providers.
- **Inheritance:** Reusable schema structures are extended in backend models.
- **Singleton Pattern:** React Context acts as a global state manager.
- **Single Responsibility Principle:** Each component handles a specific functionality.

6. Challenges & Learnings

During development, several challenges were encountered, including managing global state efficiently, handling secure authentication, and ensuring smooth communication between frontend and backend. Implementing the Context API helped in solving state management issues, while JWT ensured secure user sessions. This project significantly improved understanding of full-stack development, API design, and scalable architecture.

7. Conclusion

The SDSE Food Delivery Application successfully demonstrates the integration of modern web technologies and software engineering principles. It provides a scalable, secure, and user-friendly platform that can be further extended with features such as real-time tracking, payment gateway integration, and recommendation systems.